02 · Crime and deterrence: a county panel

This report's text is generated automatically from the code's outputs. The main output is the quantitative part: data, checks, models and tables. It extracts the quantitative work of a longer paper I wrote for a UFRJ econometrics course; the full paper is available on request.

Question

Do counties with a higher probability of arrest, conviction and prison have less crime, and how much of the answer changes once fixed county traits, common year shocks and serial correlation are handled?

Result

Specification	log P(arrest)	log police per capita
Pooled OLS	-0.633	0.352
County fixed effects	-0.395	0.422
Two-way fixed effects	-0.360	0.422

• Breusch–Pagan LM = 191.60 and White LM = 410.04, both with $p < 0.0001$: heteroskedasticity is clear.

• In the two-way model, clustering by county raises the standard error on the arrest elasticity from 0.033 to 0.064.

• On police per capita the clustered standard error is 0.093, against 0.027 before. Both coefficients keep three stars after clustering.

• Fixed effects cut the pooled arrest elasticity by about a third, so part of the pooled deterrence was county traits.

Reading: higher arrest probability goes with less crime even within a county. The positive police coefficient is the classic simultaneity sign (counties hire police where crime is high), so none of this is a causal estimate.

Data checks

10 pass, 2 warnings, 0 fail (details in output/data_checks.json).

Main tables

MQO agrupado vs. efeitos fixos de condado (within)

	MQO agrupado	EF de condado
	(1)	(2)
ln(prob. prisao) β_1	-0,633***	-0,395***
	(0,033)	(0,034)
ln(prob. condenacao)	-0,452***	-0,309***
	(0,022)	(0,022)
ln(prob. encarc.)	-0,150***	-0,211***
	(0,049)	(0,033)
ln(sentenca media)	-0,078**	0,032
	(0,038)	(0,026)
ln(policia p.c.) β_2	0,352***	0,422***
	(0,023)	(0,028)
Controles socioeconomicos	Sim	Absorvidos
Efeitos fixos de condado	Nao	Sim
Efeitos fixos de ano	Nao	Nao
Erros-padrao	Convencional	Convencional
N (obs.)	630	630
R^2	0,805	0,373

Source: output/q3_tab_fe_condado.tex

Efeitos fixos de condado vs. painel de duas vias (condado + ano)

	EF condado	EF duas vias
	(1)	(2)
ln(prob. prisao) β_1	-0,395***	-0,360***
	(0,034)	(0,033)
ln(prob. condenacao)	-0,309***	-0,286***
	(0,022)	(0,021)
ln(prob. encarc.)	-0,211***	-0,184***
	(0,033)	(0,033)
ln(sentenca media)	0,032	-0,003
	(0,026)	(0,027)
ln(policia p.c.) β_2	0,422***	0,422***
	(0,028)	(0,027)
Efeitos fixos de condado	Sim	Sim
Efeitos fixos de ano	Nao	Sim
Erros-padrao	Convencional	Convencional
N (obs.)	630	630
R^2	0,373	0,370

Source: output/q3_tab_duas_vias.tex

Painel de duas vias: erros convencionais vs. clusterizados por condado

	Convencional	Cluster
	(1)	(2)
ln(prob. prisao) β_1	-0,360***	-0,360***
	(0,033)	(0,064)
ln(prob. condenacao)	-0,286***	-0,286***
	(0,021)	(0,056)
ln(prob. encarc.)	-0,184***	-0,184***
	(0,033)	(0,049)
ln(sentenca media)	-0,003	-0,003
	(0,027)	(0,037)
ln(policia p.c.) β_2	0,422***	0,422***
	(0,027)	(0,093)
Efeitos fixos de condado	Sim	Sim
Efeitos fixos de ano	Sim	Sim
Erros-padrao	Convencional	Cluster (condado)
N (obs.)	630	630
R^2	0,370	0,370

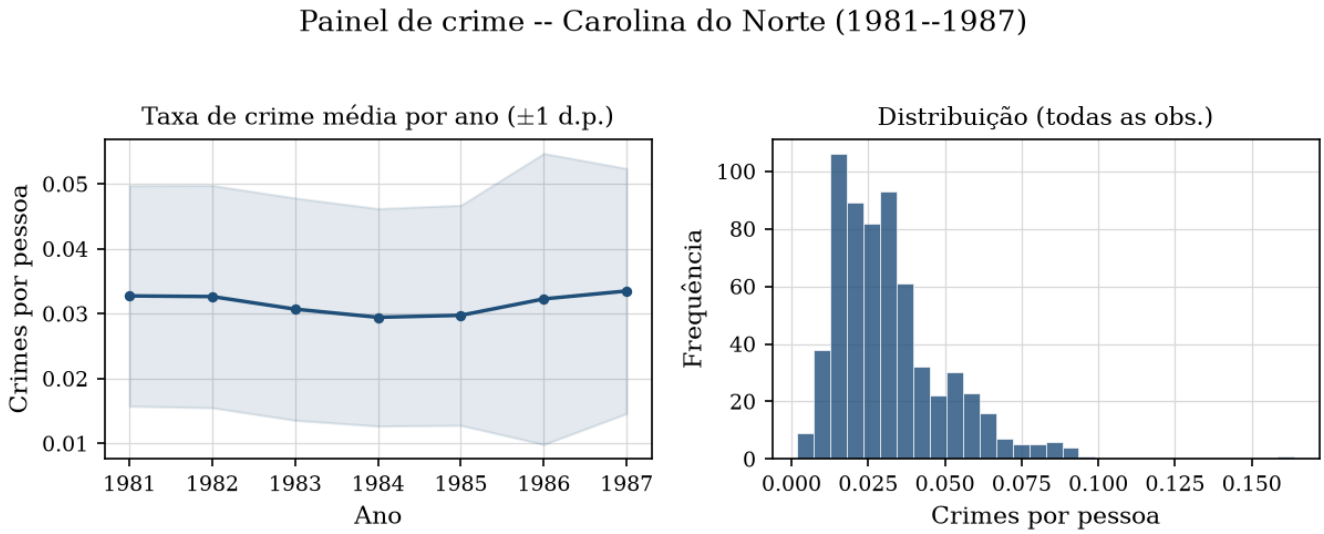
Source: output/q3_tab_cluster.tex

Testes de heterocedasticidade no MQO agrupado

Teste	Estatistica LM	g.l.	p-valor
Breusch–Pagan	191,60	14	<0,0001
White	410,04	115	<0,0001

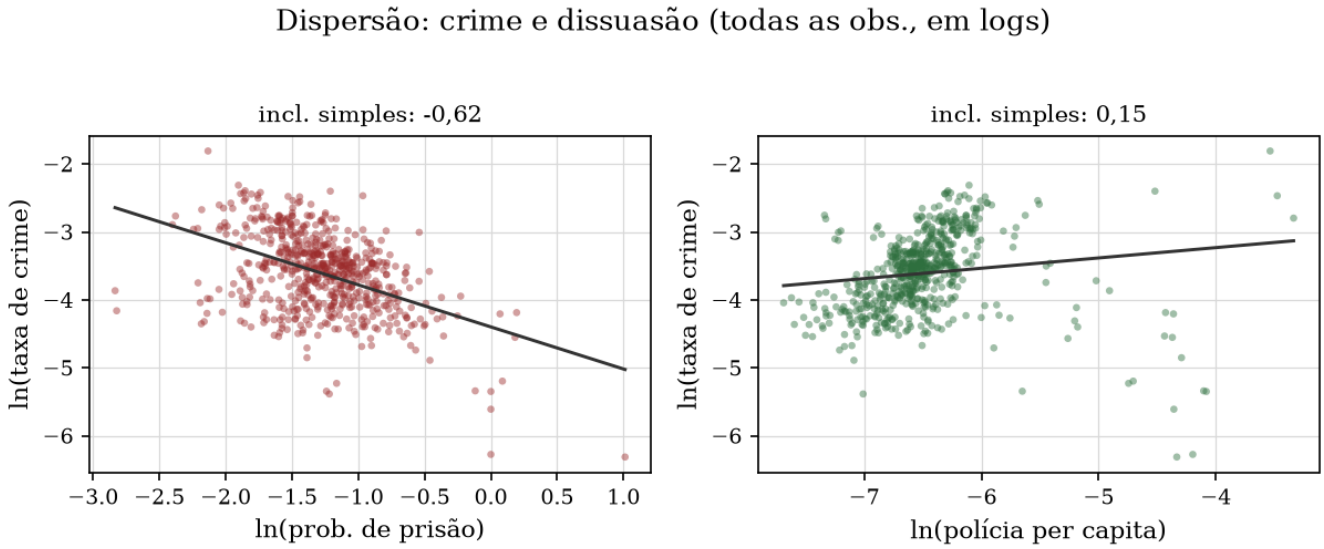
Source: output/q3_tab_heterocedasticidade.tex

Main figures



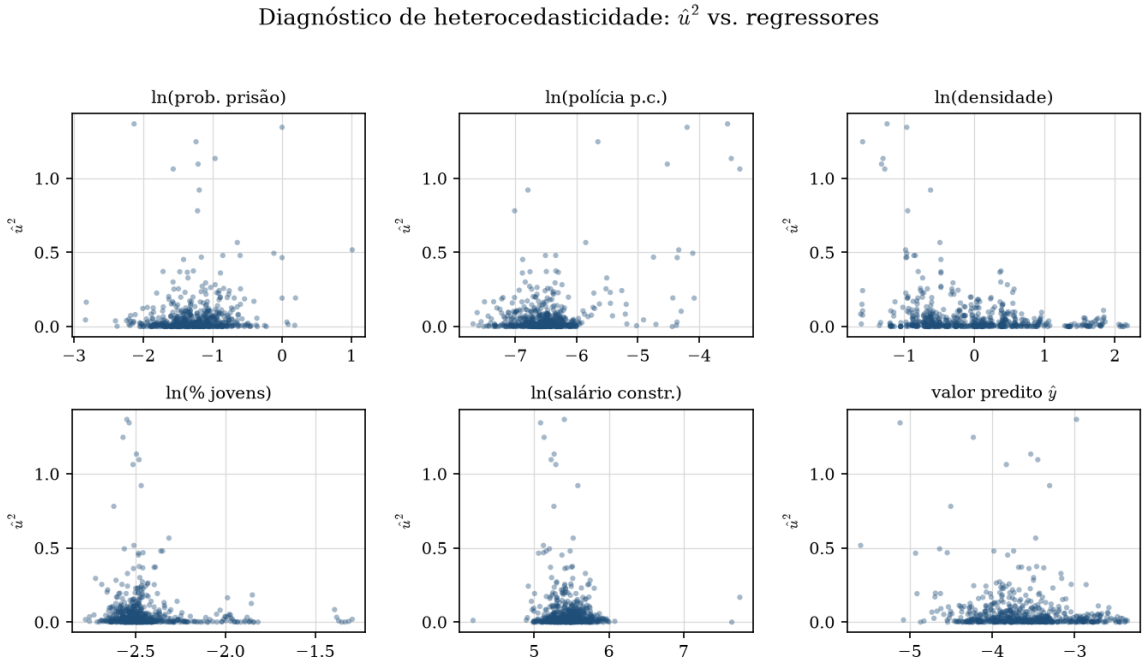
Fonte: Cornwell e Trumbull (1994), painel de 90 condados da Carolina do Norte (1981-1987), via wooldridge::crime4. Elaboracao propria.

Source: output/q3_fig_painel_descritivo.png



Fonte: Cornwell e Trumbull (1994), painel de 90 condados da Carolina do Norte (1981-1987), via wooldridge::crime4. Elaboracao propria.

Source: output/q3_fig_dispersao.png



Fonte: Cornwell e Trumbull (1994), painel de 90 condados da Carolina do Norte (1981-1987), via wooldridge::crime4. Elaboracao propria.

Source: output/q3_fig_residuos_scatter.png